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Global Markets Strategy
22 July 2026

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Introducing a cross-market framework for front-end DM rates

- The J.P. Morgan Quantamental System (JPMaQS) is a unique set of quantitative-fundamental indicators that reflect the "information state" of global financial markets at the point-in-time when those economic statistics were released.
- We explore a variety of JPMaQS indicators representing common drivers of front-end DM rates, including inflation expectations, employment, and GDP growth, and construct macro scorecards that visualize periods of overheating and slack in various economies.
- We construct a cross-market model of 2Y OIS rates, and utilize the cross-comparable nature of the JPMaQS indicators to draw conclusions about the relative reaction functions of DM central banks, and the varying weights of macro-fundamental drivers across the DM...
- ...we find that inflation expectations are the most consistent driver of front-end rates, but that the sensitivity to labor markets varies, finding the greatest impact in regions where the central bank has a dual mandate, or puts more emphasis on labor market stability.
- Additionally, this framework suggests that the majority of global front-end rates are significantly too cheap after adjusting for fundamental drivers, with the lone exception of Japan, which appears too rich on this basis.

Rates Strategy

Elisabetta Ferrara AC
(44-20) 7134-2765
elisabetta.ferrara@jpmorgan.com
J.P. Morgan Securities plc

Amanda Berke AC
(1-212) 834-5739
amanda.berke@jpmorgan.com
J.P. Morgan Securities LLC

Jay Barry
(1-212) 834-4951
john.f.barry@jpmorgan.com
J.P. Morgan Securities LLC

Francis Diamond
(44-20) 7134-1504
francis.diamond@jpmorgan.com
J.P. Morgan Securities plc

See page 10 for analyst certification and important disclosures.

Elisabetta Ferrara [AC] (44-20) 7134-2765
elisabetta.ferrara@jpmorgan.com
J.P. Morgan Securities plc
Amanda Berke [AC] (1-212) 834-5739
amanda.berke@jpmorgan.com
J.P. Morgan Securities LLC
Jay Barry (1-212) 834-4951
john.f.barry@jpmorgan.com
J.P. Morgan Securities LLC
Francis Diamond (44-20) 7134-1504
francis.diamond@jpmorgan.com
J.P. Morgan Securities plc

Intro to JPMaQS

The J.P. Morgan Quantamental System (JPMaQS) is a unique set of quantitative-fundamental indicators that reflect the “information state” of global financial markets as experienced by market participants at the time they are released, free from the future revisions typically embedded in most other macroeconomic data sources. Traditional macro data series often present hidden challenges that can impact their interpretation: a change in calculation or collection methodology might shift the entire visible backseries of data, for example, and data revisions can appear days, months, or years after first release. While this is a standard part of macro data collection, there are follow-on implications for modeling. After all, the market reaction to a particular print is based on the data as it appears at the time of release, a number that may no longer be visible or easily accessible as time passes. Calibrating trading strategies and macro fundamental models of market behavior, therefore, also becomes an exercise in data cleaning and specialized knowledge, to properly mitigate the effects of these after-the-fact updates. This problem is multiplied many-fold when constructing global models, as every local dataset and statistical agency naturally requires a unique approach.

In contrast, the indicators contained in JPMaQS represent the series as they were received by the market at the time of release, with the handling and processing of the data already embedded. This makes the indicators a ready-to-use option for modeling and backtesting. In this note, we explore a variety of the available JPMaQS indicators, use those factors to construct a simple fundamental model for front-end DM rates, and draw conclusions from the findings.

Selection of JPMaQS indicators

JPMaQS contains both point-in-time variations on more traditional/recognizable data series, and novel indicators calculated from raw data that use construction methods unique to the dataset. The methodology is kept as consistent as possible across markets, so the resultant indicators also have enhanced cross-comparability, though of course the underlying data are still subject to the typical publication lags and general collection issues inherent in the relevant national statistical agencies.

Our eventual goal is to model front-end DM rates, and more specifically 2Y OIS rates, as a function of a small set of macro fundamentals that capture the drivers of the expected policy path. To that end, we explore a range of JPMaQS indicators that capture common drivers of front-end rates—inflation expectations, GDP growth, and employment—and construct the variables in such a way that a positive value demonstrates the relative “overheating” or “slack” in each economy. In this way, JPMaQS allows us to create a loose interpretation of the Taylor rule, but using real-time, observable inputs. These indicators are:

- **Excess 1-year ahead inflation expectations:** We use the JPMaQS estimated 1-year ahead inflation expectations, which assumes that market participants form their inflation expectations based on recent inflation data (an average of headline and core inflation, with a 50% adjustment for jumps and outliers) relative to the effective inflation target (which takes the official stated inflation target and adjusts for how far off-target inflation has run over the last 3 years). We then adjust by the effective inflation target of the relevant economy to form a measure of excess inflation expectations. Note that the underlying assumption of this methodology is the

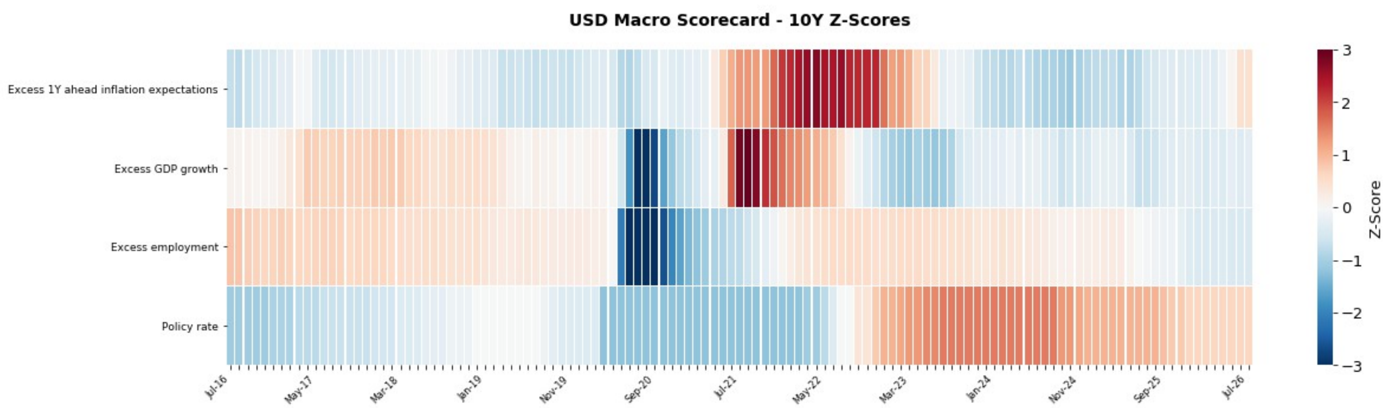
Elisabetta Ferrara AC (44-20) 7134-2765
elisabetta.ferrara@jpmorgan.com
J.P. Morgan Securities plc
Amanda Berke AC (1-212) 834-5739
amanda.berke@jpmorgan.com
J.P. Morgan Securities LLC
Jay Barry (1-212) 834-4951
john.f.barry@jpmorgan.com
J.P. Morgan Securities LLC
Francis Diamond (44-20) 7134-1504
francis.diamond@jpmorgan.com
J.P. Morgan Securities plc

credibility of the relevant CB; the formula adjusts for multiple years of off-target inflation, but does not account for sudden shifts in central bank credibility. Critically, this is not a survey-based measure of inflation, but instead anchors inflation expectations as a function of realized inflation and a gradual return to target. At times, this may yield results that appear at odds with more reactive, survey-based measures, which may respond to external stimuli not seen in realized data.

- **Excess employment:** The inverse of the seasonally adjusted unemployment gap, which is the seasonally adjusted unemployment rate, 3-month moving average minus the 5-year moving average.
- **Excess GDP growth:** This is the JPMaQS "technical" GDP growth estimate, which nowcasts GDP using vintages of standard econometric estimates. Specifically, this is the 3-month moving average of the latest estimated technical real GDP growth trend (%oya), minus a long-term (5-year) median of that country's actual GDP growth rate.

Figure 1uses these indicators to construct a scorecard of the US economy based on 10-year z-scores, visualizing periods of overheating or slack. Note that the point-in-time nature of JPMaQS means the indicators are specifically demonstrating overheating or slack a*s the market actually observed it in the data at the time*, as opposed to a revised series reflecting information gained only in hindsight.

Figure 1: We construct a scorecard of the US macro economy, visualizing periods of slack and overheating as the market observed at the time
Summary scorecard of JPMaQS US macro-economic indicators, using 10-year z-scores;



Source: J.P. Morgan

As mentioned, a benefit of the JPMaQS dataset is the uniform calculation methods used across local markets, which makes the indicators well-suited to cross-market comparison. **Figure 2** plots the z-score of the most recent observation in a 3-year period, giving a sense of more recent economic momentum. In this view, the 1-year ahead inflation expectations in the US are clearly outsized versus recent history, reflecting the continuation of over 5 years of above-target inflation, in the context of a policy rate that has been cut from more restrictive levels. In AUD, both excess employment and growth are well below 3-year averages, as the policy rate sits at more restrictive levels to match stronger excess inflation expectations. We can also observe the clear differentiation of the underlying macro story in Japan versus the rest of the DM, where the policy rate is historically high as the hiking cycle has commenced alongside above-average growth, though the excess inflation measure is below recent averages.

Elisabetta Ferrara AC (44-20) 7134-2765
elisabetta.ferrara@jpmorgan.com
J.P. Morgan Securities plc

Amanda Berke AC (1-212) 834-5739
amanda.berke@jpmorgan.com
J.P. Morgan Securities LLC

Jay Barry (1-212) 834-4951
john.f.barry@jpmorgan.com
J.P. Morgan Securities LLC

Francis Diamond (44-20) 7134-1504
francis.diamond@jpmorgan.com
J.P. Morgan Securities plc

Figure 2: We also construct a cross-market macro scorecard, contrasting overheating in various parts of the DM

Z-score of the most recent observation in a 3-year historical period of various JPMaQS macro-economic indicators;

Cross-market Macro Scorecard - 3Y Z-Scores

	USD	EUR	GBP	SEK	NOK	JPY	AUD	NZD
Excess 1Y ahead inflation expectations	2.7	1.0	-0.2	-0.4	-0.4	-1.6	0.5	0.0
Excess employment	-1.5	-2.0	-1.0	-0.8	-0.5	-0.8	-1.6	-1.0
Excess GDP growth	0.6	0.3	-0.4	-0.1	0.1	1.4	-1.6	1.1
Policy rate	-1.3	-0.8	-1.3	-1.0	-0.3	2.0	0.8	-1.1

Source: J.P. Morgan

Introducing a 2Y OIS rate fair value model for DM markets

Beyond simply visualizing these indicators, we can use them to introduce a model for 2Y OIS rates as a function of this small set of macro fundamental drivers. We include the variables detailed above to represent excess 1-year inflation expectations, growth, and employment, and include the current policy rate as well. The first three variables are expressed as deviations from a benchmark in %-pts, and measure how much pressure the macro backdrop is putting on the central bank, while the policy rate provides the level anchor, consistent with the idea that the 2Y largely reflects the current policy stance plus the expected path. **Intuitively, the model gives us a simple fair value estimate of where the 2Y should be trading, given the macro backdrop and where policy sits today**.

We expect positive coefficients across all four drivers (**Figure 3**). Higher excess inflation expectations signal more persistent price pressure relative to target, which is typically associated with a tighter expected policy path/higher 2Y OIS yields. Stronger growth and tighter labour markets relative to trend likewise reduce downside risks and raise the odds of restrictive policy, pushing front-end yields higher. Finally, a higher policy rate anchors the 2Y at a higher level by raising the starting point for the expected policy path.

Figure 3: We model front-end DM rates, specifically 2Y OIS rates, as a function of a small set of macro fundamentals that capture the drivers of the expected policy path

Variable	Expected sign	Interpretation
Excess 1Y ahead inflation expectations	+	Higher realised inflation and expected persistence versus central bank target raise the odds of further tightening/slower easing, pushing yields higher
Excess employment	+	Higher employment versus norm pushes yields higher
Excess technical GDP growth	+	Stronger growth versus trend pushes yields higher
Policy rate	+	Higher policy rate reflects a tighter current policy stance and higher starting point for expected path

Source: J.P. Morgan

We estimate a simple OLS regression of the 2Y OIS rate level on these variables over a 10-year horizon (Figure 4). We choose a 10-year lookback window on the hypothesis that fundamental macro variables tend to be slow moving. While we might typically shy away from using a longer window due to increased revisions to older data,

Elisabetta Ferrara [AC] (44-20) 7134-2765
elisabetta.ferrara@jpmorgan.com
J.P. Morgan Securities plc
Amanda Berke [AC] (1-212) 834-5739
amanda.berke@jpmorgan.com
J.P. Morgan Securities LLC
Jay Barry (1-212) 834-4951
john.f.barry@jpmorgan.com
J.P. Morgan Securities LLC
Francis Diamond (44-20) 7134-1504
francis.diamond@jpmorgan.com
J.P. Morgan Securities plc

the JPMaQS indicators lend themselves well for this purpose. The model fit is strong across all markets, with adjusted R-squared ranging from around 0.91 in AUD to around 0.97 in EUR, GBP and NZD. The strong fit was maintained even when testing the model at various shorter regression windows, and we explore those results in more detail later in this note. As expected, the policy rate is the dominant driver of front-end yields across all markets, with positive and significant coefficients ranging from 0.75 to 0.85 in most cases, consistent with the 2Y being anchored by the current stance. **Beyond the policy anchor, inflation expectations are the most consistent macro driver, carrying a positive (and significant) coefficient across markets. Growth and labor market terms add explanatory power, but are less uniform across countries.**

Figure 4: A simple OLS regression over the past 10 years shows that, beyond the policy anchor, inflation expectations are the most consistent macro driver (positive and significant coefficient) of 2Y OIS rates across DM

Statistics from regressing 2Y OIS rates (%) across the DM on selected JPMaQS variables and policy rate (units as indicated), regression over the last 10-years;

	Adjusted R2	Constant		Excess inflation expectations (%-pt)		Excess employment (%-pt)		Excess GDP growth (%-pt)		Policy rate (%)	
		coef	t-stat	coef	t-stat	coef	t-stat	coef	t-stat	coef	t-stat
USD	94%	0.11	6.5	0.35	42.5	**0.14**	21.1	-0.06	-10.3	0.75	135.2
EUR	97%	0.32	19.7	0.38	86.8	-0.17	-17.4	0.00	*-0.3*	0.76	214.4
GBP	97%	0.21	15.1	0.40	72.9	-0.09	-9.1	0.00	*1.4*	0.84	215.2
SEK	95%	0.30	31.1	0.30	62.4	-0.02	-2.6	0.05	15.7	0.77	176.8
NOK	95%	1.07	77.3	0.34	39.7	0.23	19.3	0.08	16.7	0.78	198.3
AUD	91%	0.54	22.0	0.27	21.9	0.19	11.5	0.12	10.7	0.85	108.5
NZD	97%	0.80	66.1	0.65	107.9	0.02	*1.8*	0.14	49.5	0.76	221.6
JPY	95%	0.17	62.1	0.07	30.4	-0.07	-13.3	0.00	3.2	1.47	206.4

Note: T-stat < |1.96| not statistically significant at 5% level (results shaded in grey text).
Source: J.P. Morgan

Notably, as the JPMaQS methodologies are uniform across markets, we can use the relative betas to draw conclusions about central bank reaction functions and cross-market drivers. As one would expect, the inflation variable has the highest explanatory power of the macro drivers, consistent with the fact that all eight central banks target inflation. This sensitivity is largest in NZD (0.65), which fits the RBNZ's historically aggressive inflation-targeting approach. However, divergences emerge with respect to sensitivity to employment, which is materially higher (and positive) for USD, NOK and AUD, but the inverse for GBP and EUR. Intuitively, this makes sense, as both the Fed and RBA have dual mandates, and Norges Bank has historically placed greater weight on domestic slack, versus the clear, inflation-only mandates of the ECB and BoE. Turning to growth, the largest coefficients appear in AUD, NZD, NOK and SEK, suggesting expectations for the policy path are more sensitive to the growth cycle in smaller, open economies, where central banks have historically responded more directly to external demand conditions and the broader activity backdrop.

Japan is the notable exception, with by far the largest policy rate coefficient of 1.47. We attribute this to Japan's extended ELB period, which loads explanatory power onto the policy anchor. We separately tested JPY both with and without the policy indicator over shorter lookback windows, to test the relationships in the post-lift-off period, and indeed the explanatory power of the fundamental variables improves when restricting the model in this way. However, the signs of the coefficient are not intuitive, perhaps indicating other factors (like supply expectations) could improve the fit of the model. Additionally, the policy persistence of the BoJ has been an important feature of the JGB

Elisabetta Ferrara [AC] (44-20) 7134-2765
elisabetta.ferrara@jpmorgan.com
J.P. Morgan Securities plc
Amanda Berke [AC] (1-212) 834-5739
amanda.berke@jpmorgan.com
J.P. Morgan Securities LLC

Jay Barry (1-212) 834-4951
john.f.barry@jpmorgan.com
J.P. Morgan Securities LLC
Francis Diamond (44-20) 7134-1504
francis.diamond@jpmorgan.com
J.P. Morgan Securities plc

market, with broader implications for the rates complex, and therefore we prefer keeping the policy anchor to reflect this dynamic. This exercise demonstrates an important limitation of the approach: **this simple, cross-market model is not intended to supplant more nuanced local modeling, which can account for the unique drivers in each market, and instead is constructed to optimize the uniformity (and therefore cross-comparability) of the factors, while still maintaining a strong fit.**

Alternate specifications and further exploration

Alongside the OLS model, we estimate an error correction model (ECM) to test whether the OLS residuals are mean-reverting and to quantify over which horizons deviations from fair value tend to close. While we prefer the OLS specification for general analysis, given the more intuitive interpretation and broadly similar results, the ECM gives an indication of which markets provide more conviction in the model. The results point to reliable mean reversion across markets. EUR, GBP and NZD show the fastest adjustment, with deviations typically closing within roughly two to three months. This analysis also suggests that **our macro fair value model should be taken as a lower-frequency valuation anchor rather than a timing tool**. Indeed, deviations can persist, particularly through rapid hiking or easing cycles where fair value is itself repricing quickly.

Given the slow-moving nature of the macro variables and the fact that the point-in-time construction of JPMaQS indicators mitigates concerns around methodology changes and revision bias over longer horizons, we have chosen to present the 10Y calibration as our baseline. This timeframe captures a decent range of policy regimes, producing more stable coefficient estimates. That said, reaction functions and the relative weight central banks place on different macro signals can shift over time. **Therefore, it is also worth presenting the model results over the past 5Y sample as a "current regime" calibration that captures the higher-inflation environment post-COVID,** as opposed to the previous low-inflation regime (**Figure 5**). An evident change in the 5Y regression is in the growth coefficients, which turn negative across most markets.

Figure 5: We also present model results over the past 5Y sample as a "current regime" calibration that captures the higher-inflation environment post-COVID

Statistics from regressing 2Y OIS rates (%) across the DM on selected JPMaQS variables and policy rate (units as indicated), regression over the last 5-years;

	Adjusted R2	Constant		Excess inflation expectations (%-pt)		Excess employment (%-pt)		Excess GDP growth (%-pt)		Policy rate (%)	
		coef	t-stat	coef	t-stat	coef	t-stat	coef	t-stat	coef	t-stat
USD	91%	0.64	9.9	0.21	10.2	0.20	7.1	-0.18	-14.3	0.54	24.1
EUR	92%	0.73	15.3	0.41	35.1	-0.75	-11.6	-0.09	-6.5	0.70	56.8
GBP	91%	0.96	19.5	0.29	23.4	0.08	2.0	-0.08	-13.6	0.66	54.2
SEK	93%	1.55	50.3	0.02	*1.9*	0.60	22.0	-0.14	-13.3	0.28	19.5
NOK	88%	1.48	53.9	0.26	17.6	0.03	*1.3*	0.00	*0.4*	0.64	80.3
AUD	82%	1.21	26.2	0.32	15.0	-0.08	-2.2	-0.22	-10.1	0.47	23.9
NZD	94%	0.87	30.4	0.69	40.7	-0.12	-3.0	0.12	13.4	0.72	90.8
JPY	96%	0.22	46.8	0.00	*-0.1*	-0.22	-5.9	-0.06	-17.2	1.28	95.1

Note: T-stat < |1.96| not statistically significant at 5% level (results shaded in grey text).
Source: J.P. Morgan

Interpreting the results

Elisabetta Ferrara [AC] (44-20) 7134-2765
elisabetta.ferrara@jpmorgan.com
J.P. Morgan Securities plc
Amanda Berke [AC] (1-212) 834-5739
amanda.berke@jpmorgan.com
J.P. Morgan Securities LLC

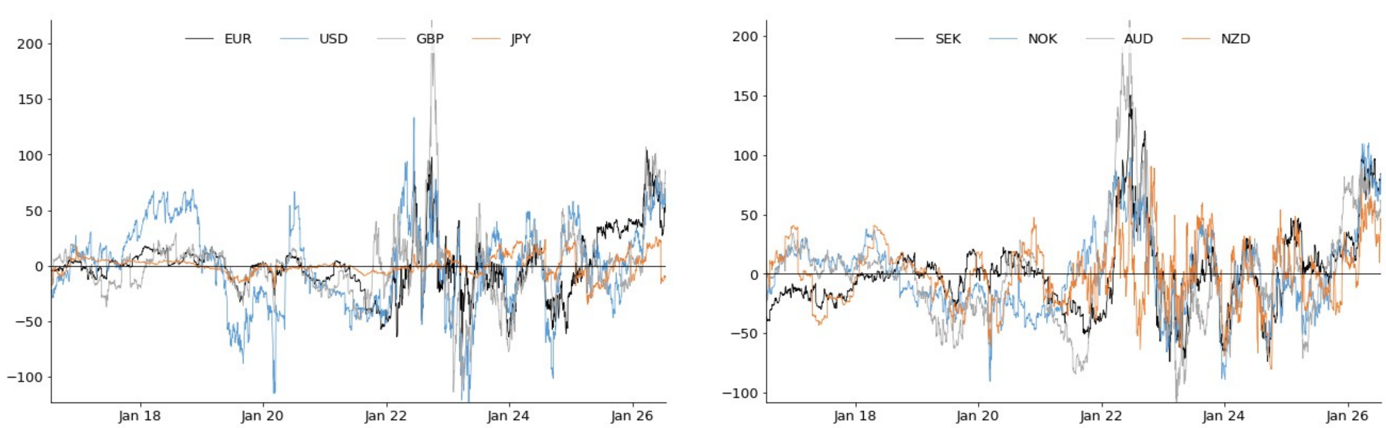
Jay Barry (1-212) 834-4951
john.f.barry@jpmorgan.com
J.P. Morgan Securities LLC
Francis Diamond (44-20) 7134-1504
francis.diamond@jpmorgan.com
J.P. Morgan Securities plc

Using a 10-year lookback window and the specification detailed above, most DM front-end rates are screening high versus macro fair value, with 2Y OIS rates pricing a more hawkish policy path than the underlying fundamental data would suggest (**Figure 6**). The divergence across most currencies opened in late 2025, but has accelerated in the wake of the conflict. This makes intuitive sense to us, as the recent rise in the front end has been driven by a quicker-moving energy shock that has not yet been captured by the slower-moving realized fundamental data. There are also regional stories that emerge from these results. In the US, for example, the front-end has screened as too cheap versus macro drivers since November 2025, around the end of the easing cycle, when the excess employment variable normalized and excess 1-year inflation expectations began moving into positive territory. Overall, this model suggests rates markets are generally pricing in a more hawkish outcome than the underlying macro economy would indicate.The lone exception is JPY, which appears nearly 11bp too low on this basis, more than the standard error of the model. This is interesting, given the general perception that the BoJ is behind the curve in normalizing policy, as administrative pressures weigh on rate decisions. The relative richness of front-end JPY yields versus their macro-implied fair value therefore likely also helps to explain the rise in long-end JPY yields and continued rise in USDJPY.

From a historical perspective, we draw confidence from the generally contained nature of the residuals in steady times — for example, the residual for JPY remained contained even during the period of negative interest rates. On the other hand, large divergences emerge in times of sharp shifts in the expected macro backdrop, consistent with the fact that the underlying fundamental drivers are slow-moving, and market expectations of the policy path may at times deviate sharply from the observed fundamental data drivers.

Figure 6: Most DM front-end rates appear too high after adjusting for fundamental macro drivers, with the lone exception of Japan

Residuals from regressing 2Y OIS rates across DM on JPMaQS variables (1Y ahead inflation expectations, excess employment and excess technical GDP growth) and policy rate; past 10Y;



Latest as of 20 July 2026.
Source: J.P. Morgan

Latest as of 20 July 2026.
Source: J.P. Morgan

While the residuals are interesting, we find cross-market comparisons are more informative on a z-score basis than the outright residuals alone, as this normalizes for each market's own residual volatility (**Figure 7**). **On this basis, GBP, NOK, SEK, and EUR stand out as the most stretched markets, with z-scores above 2, indicating the**

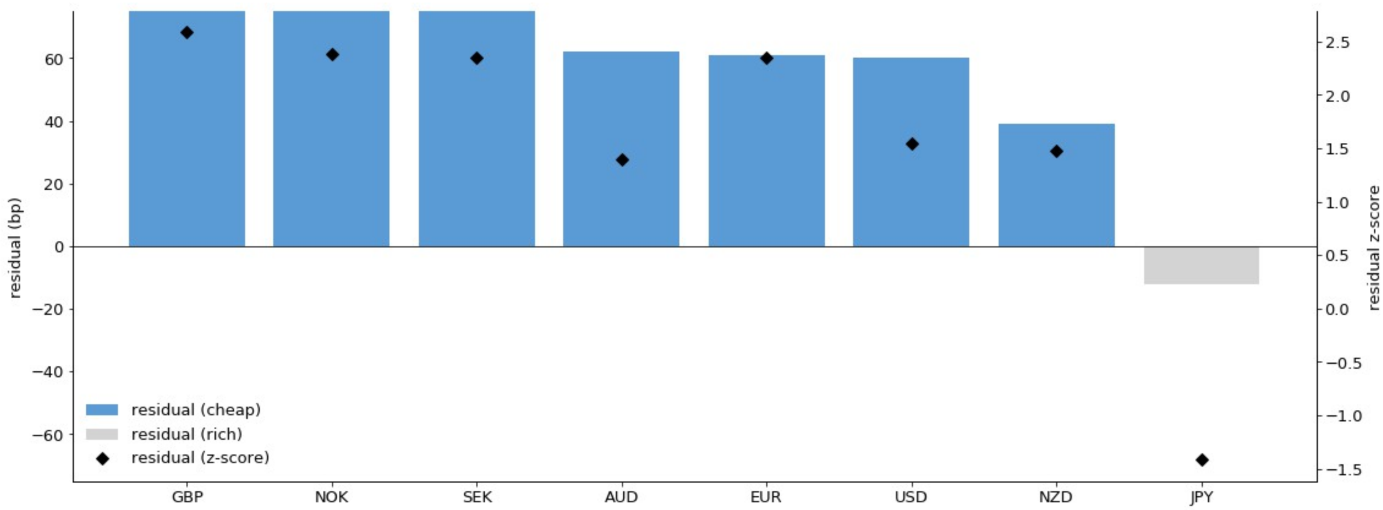
Elisabetta Ferrara AC (44-20) 7134-2765
elisabetta.ferrara@jpmorgan.com
J.P. Morgan Securities plc
Amanda Berke AC (1-212) 834-5739
amanda.berke@jpmorgan.com
J.P. Morgan Securities LLC

Jay Barry (1-212) 834-4951
john.f.barry@jpmorgan.com
J.P. Morgan Securities LLC
Francis Diamond (44-20) 7134-1504
francis.diamond@jpmorgan.com
J.P. Morgan Securities plc

expected policy path is much more hawkish than the underlying data suggests. As mentioned, JPY is the only market where 2Y OIS rates are instead screening rich versus their model-implied fair value, with a z-score of residual slightly in excess of -1.

Figure 7: Cross-market comparisons of the model residuals are more informative on a z-score basis than the outright residuals alone. GBP, NOK, SEK, and EUR stand out as the most stretched cheap markets, and JPY is the only market where front-end rates screen rich on this basis

Residuals and 5Y z-score of residual from regressing 2Y OIS rates across DM on JPMaQS variables (1Y ahead inflation expectations, excess employment and excess technical GDP growth) and policy rate; past 10Y;



Latest residuals as of 20 July 2026
Source: J.P. Morgan

On this macro-fundamental basis, the model supports being relatively overweight GBP, NOK, SEK, and EUR, and relatively underweight JPY. This broadly reflects similar trends in our more systematic frameworks. We are currently tactically long both Gilts and Bunds, and are additionally long Bunds versus Treasuries (see *Trade note: Enter longs in 10Y Germany and 10Y UK*, Francis Diamond, 7/9/26). Meanwhile, in JPY, we maintain a 5s/20s JGB curve steepener, though this is based on the timing of local supply-demand dynamics and risk factors surrounding hiking expectations (see *Japan Fixed Income Markets Weekly: Watching for signs of a GPIF Allocation Shift*, Takafumi Yamawaki, 7/17/26).

The intention of this note has been to outline the features of the JPMaQS dataset, and to investigate how the unique characteristics of the indicators can be used to demonstrate relative cross-market macro drivers of front-end DM rates. This methodology is intended as a simple, initial approach, and could easily be extended, such as by building out similar models for other rates tenors, or by investigating what types of systematic, cross-market relative value trading opportunities can be divined from the results. We intend to continue to revisit this framework in the future, harnessing the power of the JPMaQS suite to find new, compelling approaches to modeling and trade strategy.

Appendix

Elisabetta Ferrara AC (44-20) 7134-2765
elisabetta.ferrara@jpmorgan.com
J.P. Morgan Securities plc

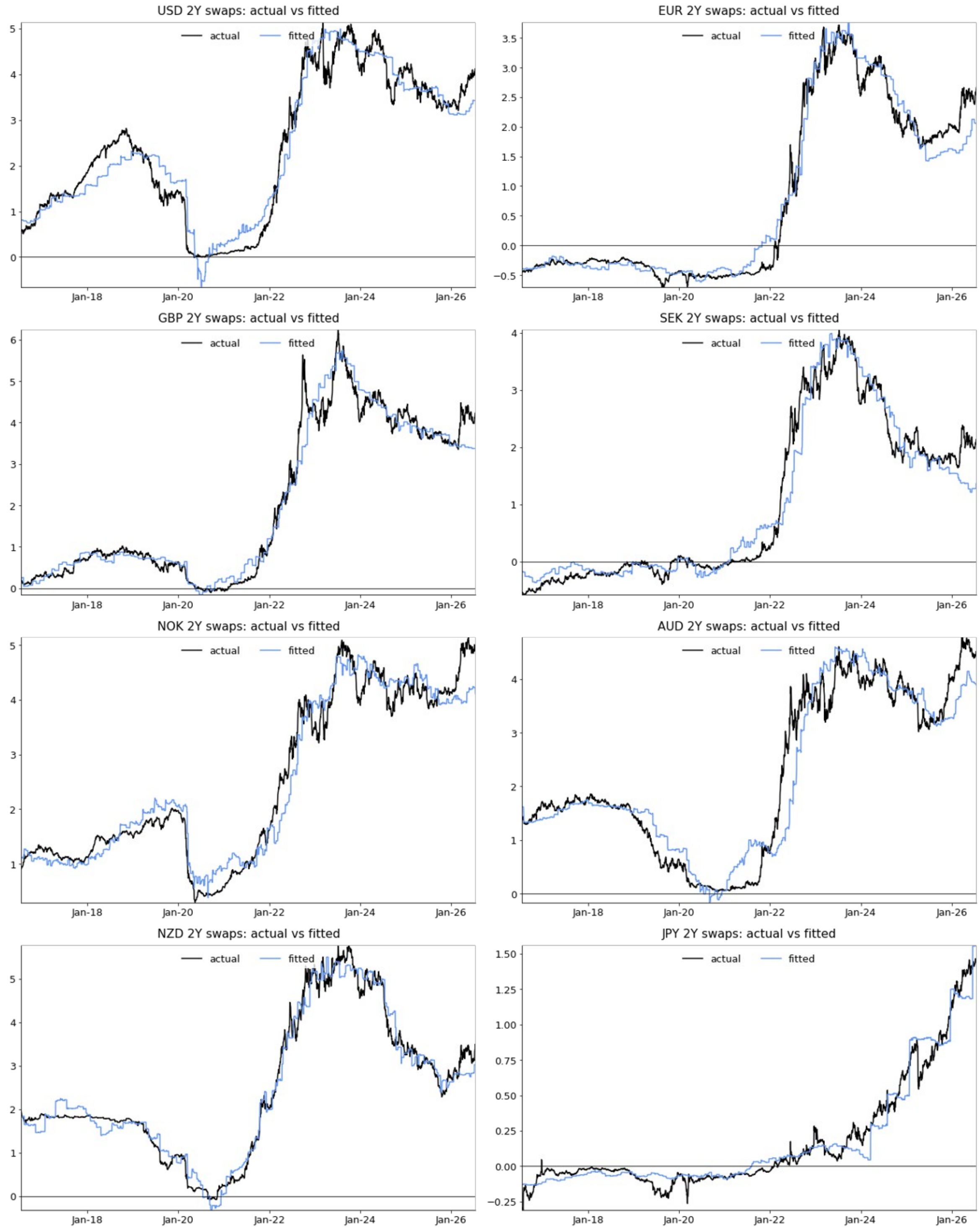
Amanda Berke AC (1-212) 834-5739
amanda.berke@jpmorgan.com
J.P. Morgan Securities LLC

Jay Barry (1-212) 834-4951
john.f.barry@jpmorgan.com
J.P. Morgan Securities LLC

Francis Diamond (44-20) 7134-1504
francis.diamond@jpmorgan.com
J.P. Morgan Securities plc

Figure 8: Currency-specific 2Y OIS rate fundamental fair value model, actual versus fitted

Regression of 2Y OIS rates on JPMaQS variables (1Y ahead inflation expectations, excess employment and excess technical GDP growth; %-pt) and policy rate across DM (%); actual vs. model; past 10Y;



Source: J.P. Morgan

J.P. Morgan Securities plc
Elisabetta Ferrara AC
(44-20) 7134-2765
elisabetta.ferrara@jpmorgan.com

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J.P. Morgan Securities plc
Elisabetta Ferrara AC
(44-20) 7134-2765
elisabetta.ferrara@jpmorgan.com

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Completed 21 Jul 2026 09:42 PM BST

Disseminated 22 Jul 2026 10:30 AM BST